

Latu Kalita

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RESEARCH INTERESTS

Theoretical condensed matter physics particularly the interplay of **topology, dissipation and non-equilibrium systems** forms the core of my research interests. I am drawn to questions connecting **non-Hermitian physics, Floquet engineering, and entanglement structure** in topological phases. Looking ahead, I aim to explore **strongly correlated systems, many-body localization, open quantum systems and field theoretic methods of condensed matter physics**.

EDUCATION

- **Indian Institute of Science Education and Research (IISER)**
Thiruvananthapuram, Kerala, India *Oct. 2025 - present*
Junior Research Fellow in a ANRF funded project.
- **Indian Institute of Technology (I.I.T) Guwahati, Assam, India** *July 2025*
M.Sc in Physics
CGPA: 8.99/10.0
- **Cotton University (CU), Guwahati, Assam, India** *June 2023*
B.Sc. in Physics
GPA: 9.43/10.0

PUBLICATIONS and PREPRINTS

- *Floquet generation of hybrid-order topology and Z_2 -like bipolar localization*, <https://arxiv.org/abs/2603.21954> (2026)— communicated in Phys. Rev. Research (under Review)
- *Floquet-Driven Superconducting Diode Effect in a Helical Shiba Chain*— manuscript in preparation.

RESEARCH EXPERIENCE

A. Indian Institute of Science Education and Research (IISER), Thiruvananthapuram

October 2025 – Present

Supervisor: Dr. Suraj S. Hegde, School of Physics, IISER TVM

- Exploring non-Hermitian topology in monitored quantum circuits, feedback-controlled topological channels, Entanglement properties of driven Kitaev-Chain, measurement-induced entanglement transitions in Majorana circuits, and emergent Fermi surfaces in non-Hermitian fermionic chains driven by skin effect.

B. Indian Institute of Technology (IIT) Guwahati

July 2024 – July 2025

Supervisor: Prof. Saurabh Basu, Department of Physics, IITG

- **Floquet-Driven Superconducting Diode Effect in a Helical Shiba Chain**
 - Demonstrated a traveling-wave drive-induced superconducting diode effect with tunable non-reciprocal critical currents and coexisting Floquet Majorana modes, requiring no external magnetic field; analyzed via self-consistent Bogoliubov–de Gennes formalism. (*manuscript in preparation*)
- **Floquet Hybrid-Order Topology and Z_2 -like Bipolar Skin Localization**

- Showed that periodic driving activates first-order helical edge states coexisting with Higher order corner modes in a (Benalcazar-Bernevig-Hughes) BBH lattice, and uncovered a drive-controlled unipolar-to-bipolar skin localization transition characterized via a non-Bloch Floquet GBZ framework.

- **Non-Hermitian Topology and Floquet Skin Effect** *(M.Sc. Thesis)*

- Studied skin effect and bulk-boundary correspondence breakdown in Su–Schrieffer–Heeger(SSH) model and 2D p -wave superconductor models using generalized Brillouin zone(GBZ) construction and Floquet formalism.

C. Cotton University, Guwahati

June 2023

Supervisor: Dr. Mridula Baro, Assistant Professor, CU

- Synthesized biomass-derived porous carbon nanomaterials via hydrothermal carbonization of bamboo powder with ZnCl_2 activation for energy storage applications.

AWARDS / SCHOLARSHIPS/FELLOWSHIPS

- **Junior Research Fellowship** *2025 - present*

Offered a position of JRF in the Anusandhan National Research Foundation(ANRF) funded project titled **Theoretical study of robust topological quantum systems and their characterization** under the guidance of **Dr. Suraj S. Hegde**, Assistant Professor, School of Physics, IISER Thiruvananthapuram.

- **Inspire Scholarship** *2020 - 2025*

Awarded by the Department of Science and Technology (DST), India, for academic excellence in science.

- **Gold Medalist** *2023*

Awarded by the Cotton University for securing First rank in the final examination of B.Sc in physics held in 2023 scoring a CGPA of 9.43 .

- **Anundoram Barooah Award** *2018*

Awarded by the Government of Assam, India for outstanding performance in the matriculation exam.

KEY COURSES TAKEN

- Mathematical Physics, Classical Mechanics, Electromagnetic Theory,Waves and Optics, Quantum Mechanics and Applications, Solid State Physics,Astronomy and Astrophysics, Thermodynamics, Statistical Mechanics, Nuclear and Particle Physics, Atomic and Molecular Physics, Quantum Field Theory, General Relativity, Quantum Information and Computation, Topics in Condensed Matter Physics.

NATIONAL EXAMS QUALIFIED

- **JOINT CSIR - UGC NET DECEMBER 2025 (Physics)** *2026*

Securing an All India Rank of 174 out of 22,533 candidates appeared , I have qualified for UGC-Junior Research Fellowship(JRF) and eligibility for assistant Professor.

- **GATE (Physics)** *2025*

Securing an All India Rank of 325 out of 19,225 candidates appeared , I have qualified Graduate Aptitude Test in Engineering(GATE) in Physics

- **IIT-JAM (Physics)** *2023*

Securing an All India Rank of 271 out of 12,740 candidates appeared, I have qualified for admission to prestigious M.Sc. programs at premier institutes like the IITs.

- **IIT JEE-MAINS**

2020

Qualified the Joint Entrance Examination (JEE-Mains) for admission to undergraduate engineering programs.

OUTREACH

- Attended 66th Department of Atomic Energy Symposium on Nuclear Physics, 2022.
- Completed a NPTEL course offered by IIT Guwahati on "Numerical Methods and Simulation Techniques for scientists and Engineers, 2021.
- Participated in the 23rd National Children's Science Congress 2015 as a Child Scientist, presenting a project titled "A Comparative Study on Locally Used Fuel (Firewood)" 2015

LANGUAGE SKILLS

- **English** : Proficient in communication. All academic degrees were completed with English as the medium of instruction.
- **Hindi** : Native Proficiency.
- **Assamese** : Mother Tongue.

SKILLS / HOBBIES

- **Programming Languages** : Python, C/C++, Mathematica, Scilab, Fortran
- **Softwares** : LaTeX, MS Word, MS PowerPoint, Excel.
- **Hobbies** : Movies, Sports and Hiking

REFEREES

- **Dr. Suraj S. Hegde**
Assistant Professor, School of Physics
IISER Thiruvananthapuram
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- **Dr. Saurabh Basu**
Professor, Department of Physics
IIT Guwahati
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